MFCC-CNN Acoustic Sensing on ESP32 for Real-Time Water Leak Detection and Cloud Notification

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Abstract—Water leakage from taps is a major source of water wastage in residential buildings, hostels, offices, and public restrooms. Minor leaks often remain unnoticed for long periods, leading to unnecessary water loss and increased maintenance expenses. Conventional leak detection methods commonly rely on moisture sensors or flow sensors, which may not effectively detect small dripping leaks at an early stage. This work presents an acoustic-based water leakage detection and notification system that continuously monitors sounds near a tap using a microphone sensor. The captured audio signals are processed and classified using a Convolutional Neural Network (CNN) model to differentiate water dripping sounds from other environmental noises such as human speech, fan noise, and running water. Upon detecting a continuous leakage event, the system activates a local alarm and sends a notification to the user through a mobile application. Leak information is stored in a cloud database for monitoring and future analysis. The proposed system provides an automated approach for early leak detection and water conservation. By identifying leakage in its initial stage and generating timely alerts, the system helps reduce water wastage and supports effective water resource management.

Keywords: Water Leakage Detection, Acoustic Signal Processing, Convolutional Neural Network, Internet of Things, Water Conservation.

I. INTRODUCTION

Water is first noticed when it is insufficient, and/or is not readily available when needed. it. But a significant portion of the water that utilities purify is not Delivers to a pay customer. Non-Revenue Water (NRW) on average is 35% in Asian cities and may reach extremely high levels, The total cost to urban utilities in Asia is estimated to be nine billion USD due to costing. US dollars a year [1]. A lot of research is dedicated to That includes the mains and service pipes of that figure. The taps But inside buildings, the attention to and fittings are far less noticed, although they are the place where most people meet the problem.

It's easy to overlook a dripping tap. Assuming roughly 0.05 mL per drop, one drop every second adds up to about four No one sees a day go by without consuming liters a day. When someone is staying in a hostel or in the public. washroom, where no one feels responsible for a tap, that drip may continue until the washer is replaced or the water bill. arrives. In fact, house leaks, in general, are common and difficult to fix. They were deemed to be at the margins of the acceptable range by spot, and studies of residential demand put them at. somewhere between 2% and 13% of total consumption [4].

The traditional means of detection are of little aid here. Flow and So pressure sensors don't measure a drip, they measure a burst. Moisture Only water that has already gathered on a is reported by sensors. surface. Using an acoustic rod to listen for the leak is not It is used by work, and is reliant on utilities, but is only accurate when the The skill and alertness of the rod holder [2, 3]. What What's required instead is something that's close to the tap, listens Stops giving the drip after it has been running for too long and notifies a person when it has been running for too long long.

This paper presents an idea for this system. An INMP441 digital microphone sends audio to an ESP32 controller, which streams the audio. to a Python service. The service fetches MFCC features and passes them to a CNN which separates leak sounds from ordinary household noise. A buzzer sounds if the drip continues on. The event is locally stored on Firebase and a notification is sent, and a dashboard shows the current status and history.

This work has the following contributions:

- with the acoustic receiver, as well as the acoustic emitter and the speech input/output devices. sensing, deep learning, IoT communication, cloud A storage and a monitoring dashboard;
- a dripping tap and a running tap by analyzing the sound.A six-class CNN which uses sound analysis to distinguish a dripping tap from a running tap. The sounds of running water, flushing, speech, fan noise are to be expected. background sound;
- a temporal confirmation rule which activates frame level Increases the accuracy of predictions to a leak event, decreasing the probability of false alarms;

- clear evaluation plan with target metrics, so that The prototype can be tested repeatedly for the same results.

The rest of the paper will be structured as follows. Section II reviews related work. The proposed The implementation is presented in Section IV, and the methodology in Section V. the signal processing and CNN. In Section VI, you will learn about leak. confirmation, alerts and dashboard. Section VII sets out the experiments and target results, Section VIII discusses The limitations and uses are outlined and the book concludes in section IX.

II. RELATED WORK

Acoustic leak detection has a long history in utilities. Vanijjirattikhan et al. built an AI-assisted system for the Metropolitan Waterworks Authority of Bangkok in which an operator records sound from a buried service pipe with an acoustic rod and a smartphone [2]. The audio is transformed with an FFT, averaged over 14 frames, normalized, and classified. Comparing SVM, DNN, and CNN on 108,481 samples from 688 leak and 619 non-leak sites, they reported test accuracies of 78.29%, 94.89%, and 94.71%, respectively. In field trials, novice operators using the device reached 97.83% accuracy against 95.56% for experts. Their work shows that a learned classifier can stand in for a trained ear, and it also shows how much care is needed when splitting data: they grouped samples by site so that one site never appeared in both training and test sets.

Several other groups have applied deep networks to leak sound. Kang et al. combined a 1D CNN with an SVM for leak recognition and added graph-based localization [5]. Müller et al., Arunsuriyasak et al., and Nam et al. worked with image-like representations of the sound, such as recurrence plots, Mel-spectrograms, and MFCCs, which CNNs handle well [6]–[8]. Gunatilake and Valls went a step further and classified the type of leak from vibro-acoustic signals in a live urban network, with accuracies up to 98% [9]. Martinez-Rios et al. compared wavelet-scattering features from accelerometer, hydrophone, and pressure data and found that an SVM on accelerometer features did best, with 89.58% and 96.67% accuracy on looped and branched test networks [10].

Closer to the present setting, Islam et al. developed a low-power IoT node that listens through a piezoelectric contact microphone, runs a quantized model of about 11 KB, and reported 98.96% accuracy; the node beeps and forwards an alert when it hears a leak [11]. Zese et al. took a different route for household plumbing and used neural networks on smart-meter consumption data to detect intra-domestic leaks of different sizes [4]. Table I summarizes how these works differ from the proposed system.

TABLE I. COMPARISON WITH RELATED WORK

Ref.	Sensing / data	Method	Setting
[2]	Acoustic rod, phone mic	FFT; SVM, DNN, CNN	Buried service pipes
[9]	Vibro-acoustic	Multiclass time-frequency CNN	Urban water mains
[10]	Accel., hydrophone, pressure	Wavelet scattering, SVM	Test pipe networks
[4]	Smart-meter flow data	Neural networks	Household plumbing
[11]	Piezo contact mic	Quantized edge ML	Home / industrial pipes
Proposed	INMP441 air mic	MFCC + CNN, temporal rule	Indoor tap, cloud alerts

Most published systems target buried pipes or large networks and rely on specialist sensors, contact microphones, or operator-held tools. A permanently mounted, inexpensive microphone that listens for tap leakage in an indoor room brings its own difficulty: the room is full of other sounds. Speech, fans, and toilet flushes all overlap with the frequencies of running or dripping water. No published work was found that combines low-cost tap-level acoustic detection, a classifier trained to reject these confounding sounds, and cloud notification with a dashboard, and this is the gap addressed here.

III. PROPOSED METHODOLOGY

The proposed approach is based on a listening to the tap via an * A microphone (INMP441) and an ESP32. The sound is cleaned up, converted to MFCC features, and classified by a CNN into six sound classes. If this drip is sustained, the buzzer sounds a warning, An alert on Firebase, and an update on the dashboard. The overall architecture is shown in Fig. 1.

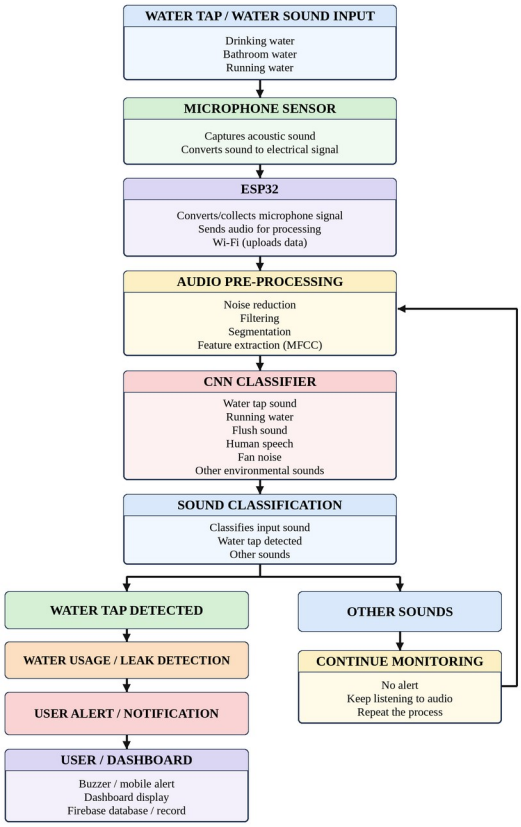


Fig. 1. Architecture flow of the proposed acoustic leak detection and notification system.

A. ESP32 Controller

The ESP32 devel board is the main controller. It has a dual-core processor, built-in Wi-Fi, low power It has reasonable consumption, and support of IoT work, low price. little. In this design it is used for three purposes: reading the microphone. through its I²S peripheral, it sends the audio to the processing If a service reports a, it will actuate the buzzer.

confirmed leak. Avoiding the significant expense of the board. The model can be retrained/replaced without means. touching the firmware.

B. INMP441 Digital Microphone

The INMP441 is a MEMS microphone based on the digital I²S interface. output. Due to the digitalization of the signal within the microphone, The analog hum on the wiring to the ESP32 is not detected by the wiring. The inexpensive analog microphone is, sufficient to hear a single fall. It is Mounted some distance from the spout and stayed away from Fans and vents should be used wherever possible. In Fig. 2, the connections are shown. The ESP32 and the buzzer are connected between the microphone.

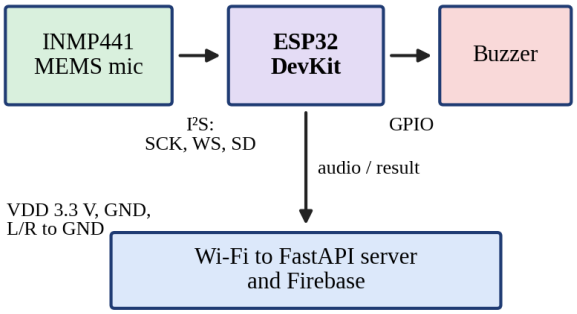


Fig. 2. Sparkfun INMP441 to ESP32 with I²S and buzzer attached to the ESP32 GPIO pin, Wi-Fi connection to the server and Firebase.

C. Buzzer and Power

A small buzzer is connected to provide a prompt local warning, which matters in places such as a washroom where the person nearby may not have a phone. When the whole node is executed from a the same node is used. controlled the 5V supply on the ESP32 board, thus allowing it to be placed Always within easy reach of the tap. For more details, refer to Processing Service and Data Flow.

D. Processing Service and Data Flow

The link between the node and the server is deliberately simple. The ESP32 has 16 bit sampling. microphones, they upload them to the FastAPI service in short. chunks over HTTP. One second of 44.1 kHz and 16 bits is equal to 18.0 MB. The audio file is approximately 88 KB, a normal WiFi connection. carries without trouble. On arrival, the service re-samples the translates the input into chunks at a 22,050 Hz sample rate and slices it into one-second chunks, Also calculates the MFCC map and is used for running the CNN. The prediction is passed on to the confirmation logic of Section VI. The service replies with a small status message, and the ESP32 switches Select the buzzer on/off as needed.

The primary output of this project is a low cost, easy-to-use Sensing node whose processing is performed solely by a separate server. There are two consequences of this arrangement. First, a new model can be trained on a laptop or server and simply entered the service, similar to the one that's The idea is the same as the The cloud-managed model updates used in [2] are applied to a Did not use a handheld device, but used a fixed node. Second, each node Assigns a location label, set at installation time, and this label travels with each record, for a notification to specify exact tap.

IV. IMPLEMENTATION

The system is developed using Python around an ESP32 node: captures tap audio. The CNN is trained using TensorFlow/Keras, and Processing and Alerts are done using FastAPI and Firebase. The software stack and dataset are described below.

A. Software Stack

Model building and analysis is conducted using Python programming. The CNN is trained with the data sets that are built and evaluated on it. TensorFlow/Keras. Librosa loads its audio files, These include resampling, MFCC extraction and spectrogram generation. [12]. The audio is sent to a FastAPI service via the ESP32 and performs the pre-processing and the model. Firebase provides The ability to synchronize with the ability to get notifications and cloud storage. the dashboard. The code is developed with VS Code. Table III lists the Identify and explain the functions of components.

B. Dataset

The data are available for the six categories of sounds listed in Table II. ESC-50 [13] provides public water. Concerning sounds that are related and sounds within the family. The remainder is given the note of At real taps, so that with the INMP441, The training data is similar to the deployed microphone's data will hear. The drips were recorded at various flow rates and at different distances, in other rooms is important, as the tap This is the case for sound, which will be shown to change noticeably with the material of the basin. room.

TABLE II. The training is carried out in sound classes.Limited to sound classes for training.

Class	Meaning	Source
Water Drip	Leak sound	Custom + public
Running Water	Normal flow	Custom + public
Flush Sound	Toilet flushing	Public + custom
Human Speech	Human voices	Custom recordings.
Fan Noise	Electrical fan	Custom recordings.
Answers to questions about the issue of background ambient noise.	Resolved ambient noise inquiries	Custom + public.

Long recordings are divided into 1-second snippets. To Keep a balance in classes and to make the model less sensitive to a particular room or microphone position, light augmentation is only added to training data: small time Changes in levels, gain changes and low-level added noise.

TABLE III. HARDWARE/SOFTWARE SUMMARY

Component	The role that a component plays in the system.
ESP32 board	Reads microphone, uploads audio, drives buzzer
INMP441	Digital MEMS microphone, I²S output.
Buzzer	Local leak alarm
Python, Librosa	A python-based framework for resampling, extracting MFCCs, analyzing them.

Component	The role that a component plays in the system.
TensorFlow / Keras	Training, evaluation of CNN using TensorFlow/Keras; Inference using CNN.
FastAPI	Receives audio, runs pipeline, replies to node
Firebase	Capable of storing data in firebase, notifications, dashboard data.Storing data to firebase, notifications, dashboard data.

V. SIGNAL PROCESSING AND CNN MODEL

A. Audio Pre-processing

Raw audio contains a lot of information which is not required. It's prepared for the network with four steps. Resampling converts All clips at 22,050 Hz, which is the sampling rate for the ESP32 recordings, The format is the same for a smartphone and ESC-50. Noise reduction removes steady background noise. Normalization normalizes the amplitude to make the model sensitive to the How loud the sound is not an issue, only the pattern of the sound. Segmentation Divides the stream into 1-second chunks. This chain is similar to the spirit of the pre-processing in [2], Pre-normalized spectra prior to classification.

B. MFCC Feature Extraction

MFCCs are the descriptors of the spectral shape of a sound on a scale of Somewhat close to human hearing [14]. Windowing each segment: To be transformed using an FFT, filtered by a mel filter bank, was converted to cepstral coefficients. The mel mapping and the coefficients are

$$m = 2595 \log_{10}(1 + f / 700)(1)$$

$$c_n = \Sigma_k \log(S_k) \cos[n(k - ½)\pi / K](2)$$

The k-th mel's energy is f/k*Sk K is the number of filters and filter is the filter. Forty coefficients are kept per frame. With an FFT size of 2048 and a hop of 512 Each one-second segment provides 44 frames from samples, a one-second segment provides 44 frames from samples, segment is now a 40 × 44 map, the CNN uses as a single-channel image.

C. CNN Architecture

A CNN suits this as a result of its filters learn small time The MFCC based frequency patterns were directly observed from the MFCC map [15]. The The broadband hiss of running water and the short isolated clicks of a steady hum, the harmonic bands of a voice, and a drip. All those “a fan”s are different in that map. The network has a capacity of two. A series of convolution–pooling blocks (32 and 64 filters of size 3 × 3). with ReLU, each followed by 2 × 2 max pooling), then a The dense layer of 128 units, dropout, and a six are added to the layer.A layer containing a dense layer of 128 units, dropout, and a six is added to the layer. way softmax output. This leaves approximately 0.92 million at the other end of the spectrum. The majority of the trainable parameters are in the dense layer.

Dropout randomly switches off units during training and Helps to prevent the network from memorizing a small data set [16]. BCE loss function during model training.The BCE loss function is employed during the training of the model. The settings for Adam optimizer [17] are given in Table IV.

TABLE IV. TRAINING CONFIGURATION

Parameter	Value
Epochs	50
Batch size	32
Optimizer	Adam
Loss function	Categorical cross-entropy
Data split	80% training, 20% testing
Input	40 × 44 MFCC map

Training segment retained and kept for a small part. This is called validation and can be monitored over the course of 50 runs. epochs. The test portion is only used once, at the end. As in [2], all The segments that are cut from the same recording session will be stored in the same One second later, next to the split, so to speak, The segments are almost the same and would otherwise raise the price of. test score.

D. Design Choices

There are several options in the pipeline that's worth considering a brief.A few options in the pipeline are worth a short. explanation. MFCCs are used instead of raw FFT This is in the form of magnitudes as in [2] since the mel scale compresses the. A compact 40-value is provided.A compact 40 value is provided for the high-frequency range. A description of each frame, which aids in maintaining small size for the network. Two convolution blocks are enough for a 40 × 44 input; a deeper stack would add parameters with little additional. The data set is expected to provide information to be learned from. modest in size. The most common default is 3 × 3 kernel. gathers a couple of adjacent mel bands over a couple of frames, which is about the scale of a single drip.

The one second segment is a compromise! It is long long enough to accommodate several drips in a typical leak, Such that the decision logic can respond in mere seconds. The dropout rate is applied only to the dense layer (value is 0.5), where almost all of the parameters are located, because where Memorisation is most likely, however. All of these values are profane. Using them as starting points, the validation set will show If they need to change, what will they change to?

VI. LEAK CONFIRMATION, ALERTS AND DASHBOARD

A. Leak Confirmation Logic

To confirm that there is a leak. One leaking segment does not necessarily indicate a leak; may only have just turned off the tap just now. The The system is thus seeking for persistence. Let $p_d(t)$ be the CNN probability of the water-drip class for segment t equal to a .Imagine that the probability of the water-drip class for segment t is a . segment count as positive when drip is the top class and its The probability of it being at least a threshold θ:

$$s(t) = 1 \text{ if argmax} = \text{drip and } p_d(t) \geq \theta, \text{ else } 0(3)$$

A run-length counter follows the positive segments:

$$r(t) = r(t - 1) + 1 \text{ if } s(t) = 1, \text{ else } 0(4)$$

A leak is detected when $r(t) \cdot \Delta t \geq T_{leak}$, with Δt being the period of time that has elapsed, and T_{leak} being the minimum time allowed. segment length. Both θ and T_{leak} are tunable per installation. The entire process is described in Algorithm 1. In

practice, a few segments of a short grace period is helpful, To ensure that one misclassified segment will not reset a long run.

Algorithm 1. Leak confirmation method

Input: Audio segment from the microphone.

Step 1: Record a brief (one second) section of the audio from the Use a microphone and perform pre-processing.

Step 2: Extract 40 × 44 MFCC map and feed to the CNN will get probabilities for each class.

Step 3: If the top class is water drip and $p_d(t) \geq \theta$, set $s(t) = 1$; otherwise set $s(t) = 0$.

Step 4: If $s(t) = 1$, update the run-length counter $r(t) = r(t - $ If $1) > 1$, then reset $r(t) = 0$ at end of grace period; else, $1) = 1$.

Step 5: If $r(t) \cdot \Delta t \geq T_{leak}$ then declare a leak and proceed to Step 6: Repeat Step 1 if it is not.

Step 6: Sound buzzer, record Event to Firebase, and notify the user of this.

Step 7: Revise the dashboard and go back to step 1 and repeat. monitoring.

Output: The buzzer can be used to make a decision on a leak, along with Firebase and the phone. alerts.

This process only begins in the drip class. Running water is What is considered “normal use” and the different treatments for flushed, speech, fan, and background? When you press sound, it takes the system back to listening. This keeps the Alarm quiet while washing/cleaning normally.

B. Firebase Storage

Once a leak is detected, the service records a leak. Firebase. The time refers to the record's time, and the location label is the location of the record. The proposed class, the model confidence and a severity level based on how long the leak has lasted. An The example is given in Fig. 3.


```
{
  "timestamp": "2026-09-28T10:35:00",
  "location": "Bathroom 1",
  "prediction": "Water_Drip",
  "confidence": 96.5,
  "severity": "Medium"
}
```


Fig. 3. Example of a Firebase event record for an actual leak.

C. Notification System

There are two types of alerts that are raised. The local alert is the buzzer, When the service reports a which the ESP32 switches on as soon as it is detected. confirmed leak. The remote alert is a firebase notification to For instance, if the user has a smartphone, the message will be sent to that device: “Water Leak Detected.” Location: Bathroom 1. Time: 10:35 AM. Severity: Medium.” The message indicates the name of the place and time. receiving it knows where to go.

D. Dashboard

The dashboard is read from the same Firebase database. It has five parts. A login and authentication module restricts Access by registered users. The real-time monitoring shows the State of each node at present time. The earlier events are listed in the leak history. With their time, place and extent. An analytics page provides monthly statistics which may indicate which taps to check. leak most often. An administrator panel allows the administrator the ability to: Manipulate nodes and users. These combined make single alarms. into information which can then be used by a maintenance team for planning.

E.4 Choosing the Decision Parameters

The balance is controlled by the threshold θ and the duration T_{leak}. The balance between fast leak detection and eliminating nuisance alarms. The results of the movement of each one are summarized in Table V. A high θ gives the system a bit of a fright, and this makes it possible to miss a slight leak. A low θ is more sensitive to small drips, allowing more noise. A short T_{leak} provides rapid warnings, but can be triggered by tapping a tap that has only just been closed. A long T_{leak} will be less prone to false alarms. but Postpones the warning. The values need to be tuned on recordings. Depending on the actual installation and varies between locations: A short T_{leak} can be used on washroom that is empty at night. A busy kitchen might require a longer one, hours while hours is adequate for hours.

The effect of decision parameters is shown in table V.The effect of decision parameters is given in table V.

Setting	Advantage	Drawback
High Θ	More false alarms	Less detection of small drips
High θ	Sends out loud drips, fewer false alarms	
Short T_{leak}	Fast alert	A warning that is triggered by a short sound.
Long T_{leak}	Very few false alarms	Delayed alert

VII. EXPERIMENTAL RESULTS

A. Evaluation Plan

The trained model is then first tested against the held-out 20% test set. It is then tested with real taps in real time using ESP32. node. The live tests include a slowly dripping tap, a tap in normal use, speaking close to the microphone, a running fan and a toilet flush. A record is kept of whether The proper class and alert conclusion were generated. Also measured is the delay between the drip crossing T_{leak} and the number of the notification sent to the phone. Long unattended monitoring false alarms.

B. Metrics

The accuracy, precision, recall and f1 score are calculated The table shows the number of true and false positives and negatives, per class and on. average. The percentage of all segments that are classified. correctly. Precision is the percentage of accurate predictions out of all the positive predictions. correct and recall is the proportion of actual positives that are found. The F1 score is the harmonic mean of precision and It is high only when both are high; recall is used to remember.

The interesting thing is that a confusion matrix is also reported, as errors are for the particular class pairs, such as running Water and fan noise.

C. Expected Results

Areas of the body based on design and on results reported for related The system is supposed to achieve the targets in Table when classified. VI. The Target column lists the expected values, and Measured column gives the values obtained for the CNN Train a model on the held out test set (Figs. 6 and 8). The false-alarm The entry level is drip-class precision 1.00, followed by The one second segment length is represented by alert delay.

TABLE VI. Identifying targets and measuring performance.

Metric	Target	Measured
Accuracy	90–95%	96.97%
Precision	> 90%	96.97%
Recall	> 90%	98.04%
F1 score	> 90%	97.29%
Alarm false rate per day (0: No false leak alarms)	Minimal	0
Alert delay after T_{leak}	Near real time	≈ 1–2 s

The CNN and DNN in [2] reached approximately the same level of performance for orientation. 94.7% - 94.9% on a 2-class leak versus no-leak task, and the edge model in [11] reported 98.96%. The present task has The 90–95% range is for six classes and a more noisy environment. Treated as realistic instead of pessimistic. The largest Confusion will also be caused by the running water, From very faint drips against background, fan noise. sound.

The results of the prototype are shown in the following preliminary figures: 4 to 8. Figs. The recording-analysis page of 4 and 5 are shown. web interface. User uploads an audio file (WAV, MP3, M4A, OGG, AAC, FLAC, or WEBM, up to 10 MB) and presses Detect leak. The page replays the song, draws a Verify that the sound-level curve is correct and returns a verdict, a confidence bar, and one line justification. Figs. 6-8 compare the two trained The models are evaluated on the evaluation data. These runs are used on uploaded. Recordings must not be captured in real time by ESP32 node, but must be captured by other devices or the internet and then uploaded to the ESP32 node for storage. Note that the false alarm and alert delay entries in Table VI should be This has been confirmed by extended live testing at actual taps.

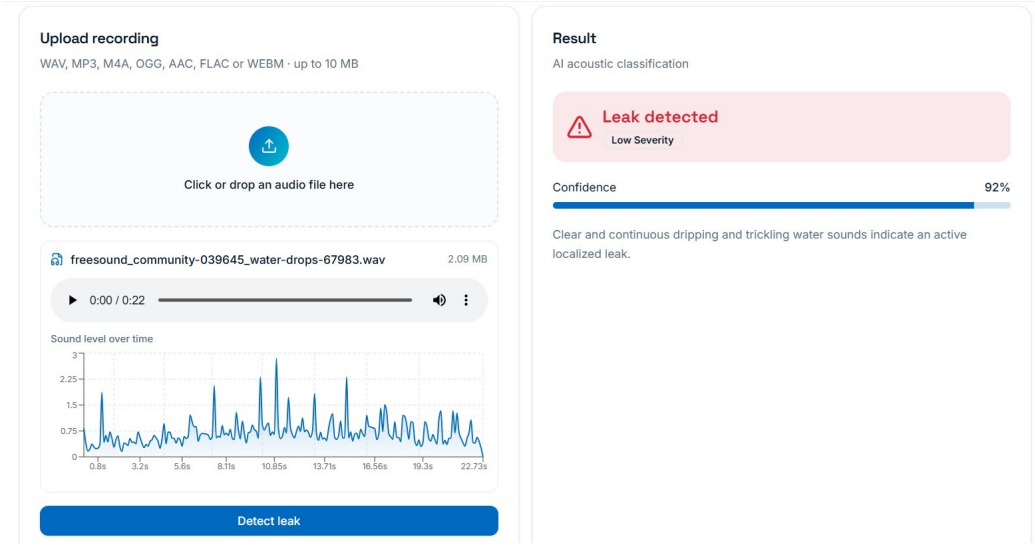


Fig. 4. The analysis page of a dripping water recording (22.7 s clip, 2.09 MB; x axis: time in s, y axis: loudness). The sound-level graph depicts sharp and isolated The peaks are read and the result panel states “Leak detected”, with 92% confidence, also low severity.

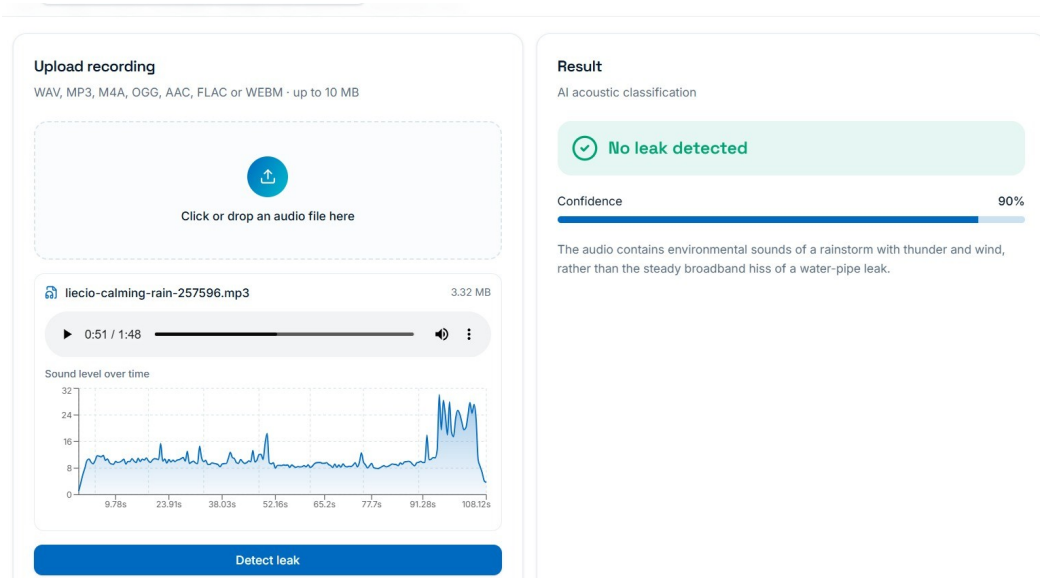


Fig. 5. Analysis page of a rain recording (1 min 48 s, 3.32 MB; x-axis: time in seconds s, y-axis: loudness). The graph depicts a gradual trend and a significant spike at the end.The graph features a gradual increase followed by a large spike at the end. At the end, the result panel reports the result as “No leak detected” with 90% confidence.

Understanding sound level graphs. In both figures the The horizontal axis (x-axis) measures time (in seconds) from The y-axis (up and down), which is the vertical axis, is the beginning of the recording. The loudness of a sound is a relative measurement, it increases with the strength of the sound. gets stronger. At each point on the curve, the loudness at the point on the curve is found. one moment – interface displays the actual time and volume level When the cursor is hovered over the curve, then value... 9.98 at 33.14 s in Fig. 5. In Fig. 4, the curve remains low, and is almost entirely ranges from 0.25 to 0.75 and is interrupted by thin peaks, and is rise to between 1.5 and 2.8. The two tallest, about 2.8 and 2.3, fall near 11 s and 15 s. One drop hits the peak, each peal is one drop hitting every peak. basin. The peaks are isolated, irregular and askew. The signal is a train of short periodic signals, repeating through the entire 22 s. impulses, rather than a regular hiss. This is the pattern the have a positive length; drip segments that are not empty; drip segments that are not disconnected; and drip segments that do not create any gaps. Repeat until $r(t)\cdot\Delta t$ is equal to T_{leak}. As the peaks are still going up, It's a leak all the way through the whole clip, until its end. Severity depends The second clip (22) is an example of how many minutes a leak has been present (prior to Section VI-B). s will be expected to be at Low.

The rain record shown in Fig. 5 is not the same. For the first The level is nearly flat at approximately 8 to 12 with only slight variations at 90 s. bumps that stay below about 18. A flat plateau is a flat terrain that is level with the sea at its base. many small drops overlap, they can't be distinguished, A continuous spectrum of sound. The level rises to near 91 s near 91 s. Between 24 and 30 and continues high until approximately 105 s prior to it. falls away. This is a thunderous burst, like the wind and thunder that were given the name. in the result text. The system reports no leak with 90 % confidence, since the sound is ambient noise and not the isolated clicks of a drip.

The two graphs also reveal that loudness is not an decide the result. The y-axis of Fig. 4 ends at 3, while that of The rain in the clip is many times louder than in Fig. 5 which reaches 32. It is the drip clip that is flagged, but not the drip clip. This agrees with the design in section: V-A, amplitude is normalized to fit the pattern of the model's response to Sound and not to its level. The two curves should therefore: Should not be compared solely on height.

Metric	MFCC Model	CNN Model
Accuracy	92.42%	96.97%
Precision	94.58%	96.97%
Recall	91.98%	98.04%
F1 Score	92.95%	97.29%

Fig. 6. The overall comparison of the MFCC model and CNN model is shown. There are four measures of accuracy, precision, recall, and F1 score.

The comparison of the two models is shown in Fig. 6. The CNN is ahead on all metrics: accuracy 96.97% versus 92.42% and precision of 100%. of 96.97% against 94.58%, recall of 98.04% against 91.98%, and F1 score of 97.29% against 92.95%. The highest difference can be found in The gap that really counts is recall – remember, about 6 points. here, since unkown leakages are not reported. The gap in The precision is poorer, on the order of 2.4 points. Both models meet the The accuracy of their precision, recall and F1 score targets, as shown in Table VI, CNN is also outside the 90-95% range of accuracy. Both are trained in such a way that the decision of which CNN is used is based on a comparison and is not assumed.

Class	Precision	Recall	F1
Background Noise	1.00	0.80	0.89
Fan Noise	0.88	0.78	0.82
Flush Sound	1.00	1.00	1.00
Human Speech	1.00	1.00	1.00
Running Water	1.00	1.00	1.00
Water Drip	0.80	0.94	0.86

Interpretation

- Water Drip Recall = **0.94**
  - 94% of actual leak sounds were detected.
- Water Drip Precision = **0.80**
  - Some non-leak sounds were wrongly predicted as leaks.
- Overall Accuracy = **92.42%**

MFCC model performs well but has a few false detections.

Fig. 7. Per-class precision, recall and F1 score of the MFCC model (overall). accuracy 92.42%).

The result of per-class MFCC model is listed in the Fig. 7. Flush The sound of a running water, human speech and sound are classified. Recall, precision and F1 score were all 1.00, perfectly. The precision of background noise is 1.00 and recall is 0.80. 20% of the background segments were re-labeled. The precision for Fan noise is 0.88, the recall is 0.45. It has an 0.78, a 0.82 and it's one of the confusions expected in Section VII-C. For the leak class, water drip has recall 0.94 but precision 0.80. So 94% of the actual drips are Although it has been found, there are also some non-leak sounds that are also referred to as leak. In a deployed node, would be displayed as false alarms. The F1 The drip class' score is 0.86.

Class	Precision	Recall	F1
Background Noise	1.00	1.00	1.00
Fan Noise	0.82	1.00	0.90
Flush Sound	1.00	1.00	1.00
Human Speech	1.00	1.00	1.00
Running Water	1.00	1.00	1.00
Water Drip	1.00	0.88	0.94

Interpretation

- Water Drip Precision = **1.00**
  - Every leak prediction was correct.
  - No false leak alarms.
- Water Drip Recall = **0.88**
  - 88% of actual leak sounds detected.
- F1 Score = **0.94**
  - Excellent balance.

Fig. 8. The CNN model results in a per-class precision, recall, and F1 score (overall). accuracy 96.97%).

The same breakdown is provided for the CNN in Fig. 8. Ambience, flush sound, voice, and running All the water samples score 1.00 on the three measures. Fan noise has recall 1.00 but precision 0.82 (F1 score 0.90), so no fan Some segments of other classes are still present, but segment is missing. labelled as fan. The only other class that has recall is water drip. All of these incorrect labels seem to be from below 1.00. missed drips are normally faint, and are interpreted as fan noise. Water The precision and recall of drip are 1.00 and 0.88 respectively, which translates to an F1 of 0.94. score of 0.94. All the leaks were right on target, There was no false leak alarm and 88% of the actual drip segments were discovered. were detected.

The CNN is a model that sacrifices a degree of model fidelity in exchange for a simple, efficient network. The percentage of little drip recalls (0.88 versus 0.94) and a large gain. of drip precision (1.00 against 0.80). This is the more favorable exchange. For a monitoring node with the primary concern of false alarms. The lost recall is partly covered by the confirmation rule of Since a leak is called after a series of positive runs, a negative run does not cancel out a previous positive run. Segments and grace period will enable a few missed segments. in that run; at most a little later. These The CNN is used in the final system, for which results are provided. Longer The delay and false alarm issues will be confirmed on live tests at actual taps. counts of Table VI.

The results in Figs. The graphs for 6 to 8 are also shown. 9 and 10, which make the differences between the two models easier to see.

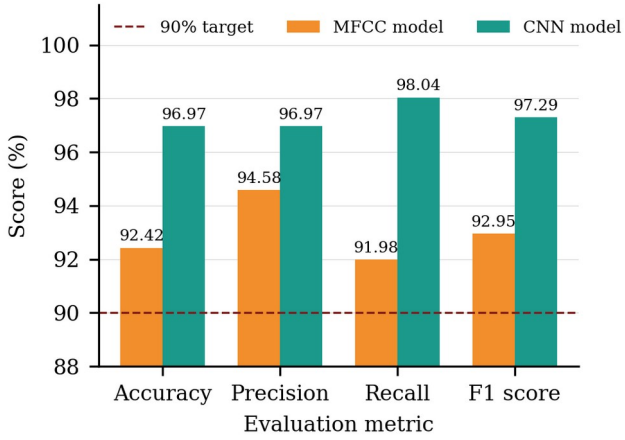


Fig. 9. The overall accuracy, precision, recall and F1 score of the MFCC Models: (orange) and CNN (teal) (y-axis is on scale starting at 88%).

Fig. 9 shows the overall metrics of Fig. 6. The x-axis lists the four evaluation metrics and the y-axis indicates the score for each one. percent. The values on the y-axis begin at 88% and do not begin at zero so the gaps between data points appear larger than they actually are. The sizes of the objects appear bigger than they actually are, the printed sizes are accurate. The CNN bars are higher on all four metrics and both models lie above the 90% target line, the MFCC model by a narrow margin in accuracy (92.42%) and recall (91.98%).

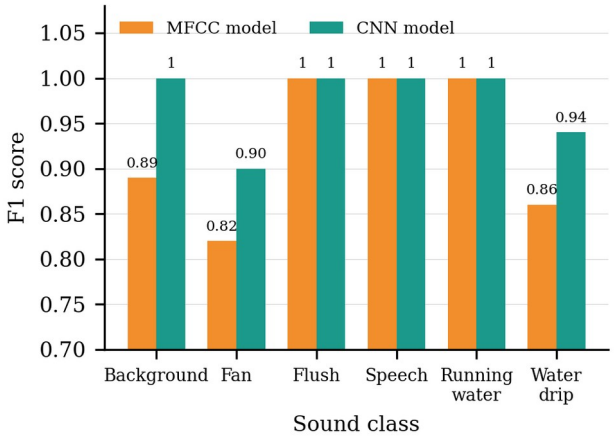


Fig. 10. F1 score per class of the MFCC (orange) and CNN (teal) models.

The F1 score for each of the classes is shown in Fig. 10. The x-axis lists The six sound classes are on the x-axis, and the y-axis is for the F1 score from 0.70 to 1.00, where 1.00 means no errors for that class. Flush The frequencies of sound, human speech, and running water are all 1.00. The difference is just in three classes, models. The CNN improves background noise (0.89 to 1.00) and water drip (0.86 to 0.94), while fan noise rises from 0.82 to 0.90. Fan noise stays the weakest class for both models, in line with the Confusion is to be expected in section VII-C.

D. Test Scenarios

To check the whole chain, and not only the classifier, the scenarios in Table VII will be run. Each has an average class and a predicted response from the system. A scenario passes only if both the classification and the alert decision are right, Note that an alarm should not go off in case of speech near the tap, for example. If it is running at the time, do not turn it off.

TABLE VII. LIVE TEST SCENARIOS

Scenario	Expected class	Expected response
Slow drip, continuous	Water Drip	Alert after T_{leak}
Normal use	Running Water	No alert.
Toilet flush	Flush Sound	No alert, YES
Talking	Human Speech	No alert - is the same as Talking Human Speech No alert.
Fan running	Fan Noise	No alert
Quiet room	Background	No alert
After the leak, drip while talking	Water Drip	Alert.

E. Threats to Validity

These three factors may bias the results to appear more favorable than they are. are. For the first time, public clips like those found in ESC-50 were not This dataset was recorded with INMP441, so a model trained primarily from it will be used. Because they might act differently on the actual device, it is important to note that; Recordings are taken at actual taps. Second, neighboring It is quite common for segments of the same recording to be quite similar, and it is for this reason that they are quite similar. the split maintains sessions. Third, a test set is prepared using only one set of scores. from the lessons you would learn in the rooms you were trained in would do little to describe a new building. Where feasible, therefore, part of the test data will be recorded in a place which the model has not heard.

VIII. DISCUSSION

A. Advantages

The design is capable of detecting a leak at an early stage, when it is still only a drip. It requires a microphone, a microcontroller and a But the hardware isn't expensive, as it is called buzzer. The AI classifier allows the ignores speech and fans, and the cloud link gives The ability to receive alerts in real-time, to view a history, and to use this information to inform their planning. maintenance. The approach is described in Table VIII. conventional options. These features assist in the support of water together. conservation directly, by shortening the time between a leak It's starting and a guy fixing it.

TABLE VIII. Comparisons are made with traditional approaches.Comparisons are made with conventional approaches.

Success is achieved through approach to	**Small drips and**	**Continuous**	**Skill needed.**
Flow sensor	Often missed	Yes	Low
Moisture sensor	After water pools	Yes	Low
High manual	No, auto No.No, high manual	No, acoustic	Yes No.
Proposed	Yes (target)	Yes	Low

A. Questions and solutions

This technique has definite bounds. There can be a very light dribble present. Close to noise floor, and a loud room can obscure it. The location of the microphone alters its sound reception, since There may be some fine tuning required on each model for each site. Because audio is sent to a server, the node depends on the Wi-Fi connection. First and foremost is the privacy, which is the microphone in a A microphone will always be used in washroom to record speech. The processing Only features and predictions should then be retained in a service's service should, Make sure to discard raw audio immediately after MFCC extraction, Make it very clear to users that the device is listening. Last but not least, a tap totally left on. open is a wasteful use but it is considered as normal running water in Provide the logic for the present case; otherwise, use a separate long-duration rule for that case. natural addition.

C. Cost and Energy

The bill of materials is quite simple: a microcontroller board, a Small microphone breakout and a Buzzer. The server side can easy to run on any small machine, or a shared cloud instance that is served by a lot of nodes at once, the cost per tap decreases as More taps are shaded. The node uses the power supplied by a Uses a mains powered adapter and doesn't require battery maintenance, Despite its Wi-Fi radio capability, it is not a great choice for battery-powered. operation. Transferring the inference to the ESP32, as suggested in Section IX, would keep the radio largely unused for the majority of the time. would have paved the way to battery use.

D. Applications

System is suitable for residential buildings and smart homes, We offer accommodation in hostels, offices, educational institutions, shopping. On multi-access facilities or public washrooms, that is, any place where: complexes are found, There are lots of taps and no one's looking at them.

E. Deployment Considerations

There are a couple more decisions to make when it comes to the practical deployment. The The distance of the microphone from the speaker should be reproducible. Ensure that the level recorded is not altered from day to day from spout spilling. day. Each node should have an identifier and a readable location name. If there are multiple nodes on a single server, the service Each node should have its own run-length counter, that is, a drip at one tap is never mixed with speech at another. Lastly, new recordings from the field, after a person Confirming if a leak was real or not, can be added to the training, set and used to retrain the model with time, as above The authors in [2] used repaired-pipe results to relabel the data.

IX. CONCLUSION AND FUTURE SCOPE

This paper presented an acoustic leak detection and notification system for taps that joins an INMP441 The microphone and controller (ESP32 with MFCC features), CNN classifier, a persistence-based confirmation rule, a buzzer, and Firebase-powered alerts and dashboard. By listening to sounds of the drip and ignoring familiar household sounds The system will attempt to provide low-cost early warning of sounds. Fluids that leak past small openings and moisture sensors are not detected by these. The next step is to increase the live tests at real taps and to validate the The measured values in Table VI.

There will be a number of extensions, scheduled for later this year. Running a compressed The transfer of the model directly to the ESP32 using TinyML would eliminate the need for additional hardware. Would be useful to stream audio, help privacy. A A specific mobile application can be used for the generic push notification. A smart valve could shut off the supply to the building if it detects a dangerous level of water. When a leak has been confirmed, it is done automatically. Beyond a single tap, the The same design can be applied to several areas of a building, use the Storing history for predictive main

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